Low-Frequency RoPE Subspace for Near–Far Sparse Attention

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***Abstract*—Page screening in sparse attention can discard useful distant tokens before refinement, making its scoring representation a critical part of candidate selection. We study a near–far cascade whose page screening and token refinement share 32 low-frequency RoPE coordinates, arranged as 16 complete rotation pairs. Separate regional page quotas retain candidate access to near and far history, while final attention uses the selected original-dimensional keys and values. In an archived configuration with 128-token prefix and recent protection, on five LongBench tasks with Qwen3-8B, the shared-subspace configuration uses 32 active coordinates for both page and token scoring and averages 47.588. Nominal variants with full-dimensional page scoring (128/32 active coordinates) or token scoring (32/128) score 47.910 and 47.666, respectively. The available records do not establish matched configurations, so these score differences are descriptive. A separate synthetic study reports no selector-latency reduction for the shared-subspace configuration relative to the nominal page-scoring variant. The results motivate evaluating compact scoring jointly with selection quality and preparation-inclusive cost, without establishing a joint quality–latency advantage.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context sparse attention must retain useful history while making selection cheaper than the attention it removes. In a two-level selector, page screening determines which tokens reach refinement; the next stage cannot recover a rejected page. Scoring dimensionality controls arithmetic per comparison, while preparation, ranking and memory access also contribute to latency. We study scoring representation and candidate allocation as distinct design choices.

We study three connected choices: shared scoring coordinates, regional candidate quotas, and an optional prefill-informed far gate (Fig. 2). The default near–far cascade uses the first two; the gate is a separate extension. Each layer and KV head uses one set of complete RoPE coordinate pairs for page screening and token refinement. The cascade assigns separate page quotas to recent unprotected history and the remaining history, allowing distant candidates to enter refinement even when the near region dominates page selection. Original-dimensional Q/K/V determine attention over the resulting token set. Sharing the scoring view reduces representational duplication in the design; regional quotas control candidate access. Neither mechanism alone guarantees lower latency or unchanged task quality.

Regional attention demand motivates examining candidate allocation separately from scoring representation. We therefore ask how regional page quotas change retrieval at a fixed final token budget, whether shared compact coordinates retain useful candidates, and whether attention savings exceed preparation and selection costs. Archived coordinate diagnostics motivate this separation but do not isolate frequency from coordinate-pair structure or evaluate the actual regional scoring functions. Layerwise prefill profiles likewise motivate testing the optional gate as an approximation policy, without assuming that prefill statistics bound far-region demand during decoding (Supplementary Note S6). Motivated by ClusterKV's token/page view [7], Figure 1 shows archived dense-reference distance density and a feasible whole-bin capture diagnostic for Qwen3-8B. Regional capture differs in these observations; this does not establish selector fidelity or task quality. The available scores and local timings neither isolate stage effects nor establish a joint quality–latency advantage.

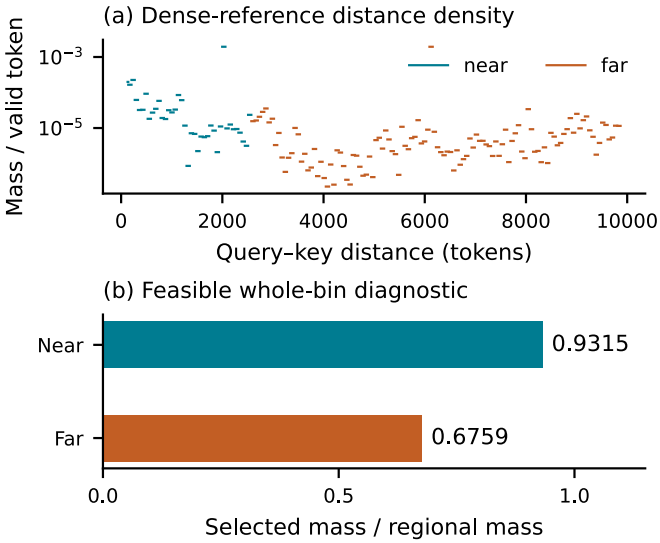


Fig. 1. Archived Qwen3-8B diagnostics, with 128/128 protection. (a) Valid-token-normalized attention mass for one MuSiQue query (sample 1, layer 24, position 9996). (b) Four source sequences contribute 128 correlated query/layer observations. Within each observation, both regional ratios use the same global whole-bin selection capped at 2048 middle tokens, with separate regional denominators. Selection acts on probabilities averaged across query heads and eight KV groups (S6). This is not the method's $K = 1024$ policy, an oracle or exact concentration.

II. Related Work

Query-aware retrieval methods such as Quest, SparQ and HISA reduce the active attention set through approximate selection [1], [2], [3]. AsyncTLS combines group-shared page/token selection, an average of per-query-head probabilities in refinement, and full-dimensional final attention [4]. Our design builds on this cascade structure and studies a different representation and allocation choice: a common set of low-frequency RoPE coordinate pairs across both stages, coupled with separate near and far page quotas. AsyncTLS instead uses full-dimensional page scoring, calibrated refinement channels and asynchronous execution. These distinctions separate the representation question from the benefits of a particular execution schedule.

KV compression and placement methods address complementary memory-capacity and data-movement constraints [5], [6], [7]. The method studied here preserves original keys and values and changes their selection. Consequently, it does not inherit a reduction in original-KV cache capacity from the compact scoring representation.

III. Near–Far Selection in a Shared Subspace

Figure 2 separates shared-subspace scoring from original-dimensional attention. The page-screening stage (L1) reserves near and far pages; the token-refinement stage (L2) ranks tokens using the same coordinates. Selected token indices and protection define the original-K/V attention set, with a separate original Q for each query. Regional quotas govern candidate access, while ordinary L2 computes full-history scores before masking.

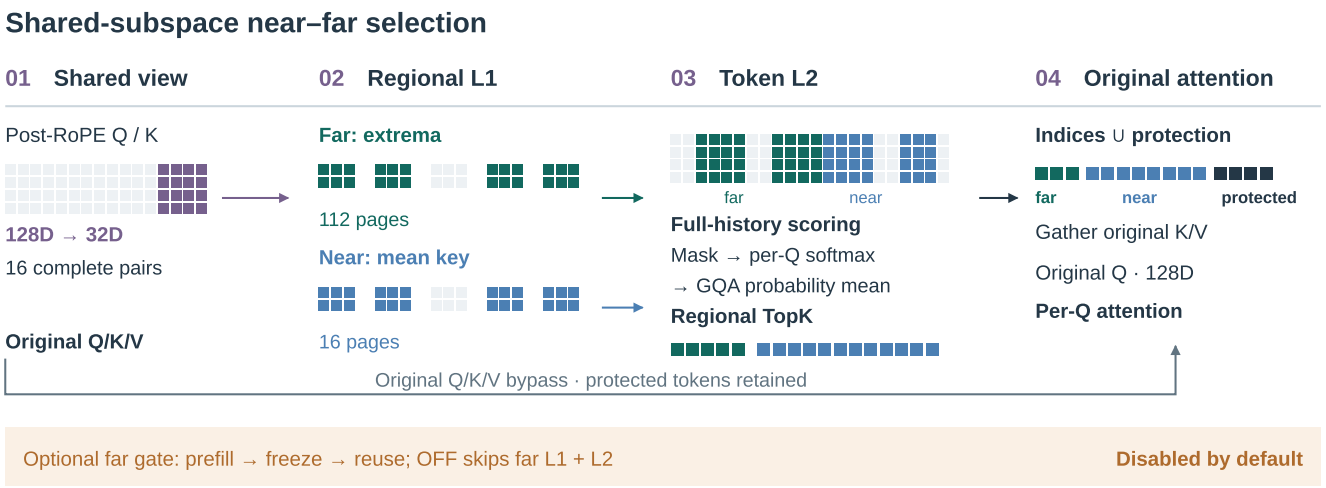


Fig. 2. Shared-subspace selection and the original-Q/K/V bypass. L1 reserves regional pages; L2 masks full-history scores, normalizes each query over shared retained support, averages probabilities within a KV group, and selects regional tokens. When OFF, the optional gate skips far scoring and redirects unused L2 quota to near candidates. Counts illustrate the sufficient-capacity configuration.

A. *Shared low-frequency coordinates*

For Qwen3-8B, the evaluated representation selects post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. These are 16 complete low-frequency rotation pairs, rather than a contiguous suffix or a learned projection. Their slower positional phase variation motivates examining them as a compact scoring view; it does not imply greater retrieval fidelity, which depends on the query and key content. The current shared-subspace implementation stores 32 coordinates compactly; the archived LongBench implementation instead used 32 active coordinates in 128-wide L2 tensors (Supplementary Note S7). With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring views are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. A layer and KV head prepare one key view for both regions and both stages (Fig. 2a; Supplementary Fig. S1). Query heads retain their own scores. Original keys and values remain available at full width for final attention and later queries; this design does not reduce original-KV cache capacity.

For key matrix $\mathbf{K}$ and row selector R, a fixed token-independent map satisfies $R(\mathbf{K}E_J) = (R\mathbf{K})E_J$: both regions can reuse its coordinates. Page summaries and quantize–dequantize token keys remain distinct derived views. Appends, region boundaries and coordinate-map changes require maintenance of those views and logical token indices. This maintenance and backend-dependent KV access determine whether reuse reduces execution time.

B. *Regional page and token budgets*

During decoding, let t be the current token's zero-based position in the full sequence, including prefill tokens, so that $S = t+1$ counts valid cached positions, including the current position and excluding padding. Let s and w denote the protected prefix and recent-window lengths. Their union and the unprotected middle are

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t - w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of the unprotected middle history; far is its complement. Near is therefore distinct from the protected sliding window. The near boundary follows fraction α of unprotected history under minimum-page and alignment rules.

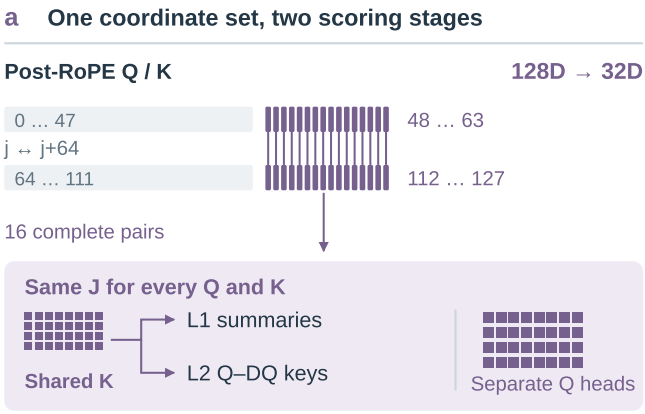


Fig. 2a. Sixteen complete split-half RoPE pairs form the 32-coordinate scoring view. Each layer and KV head derives L1 page summaries and quantize–dequantize L2 keys from the same coordinates; query views remain separate. Original Q/K/V bypass these derived views (Fig. 2).

The unprotected middle has length $\max(0, S - |\mathcal{P}_t|)$; the selected configuration is specified in the experimental setting.

Figure 2b details the regional cascade. Near uses average-key page scores. Far uses sign-aware extrema in the shared space. For each nonempty region-intersected page b, define $m_{b,j} = \min_{i\in b} z_{i,j}$ and $M_{b,j} = \max_{i\in b} z_{i,j}$. Quest's extrema construction gives

$$U_b(z_q) = \sum_j (z_{q,j}^+ M_{b,j} + z_{q,j}^- m_{b,j}), \tag{3}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. This bounds unquantized shared-space dot products, not original-dimensional logits or separately quantized scores. Extrema can come from different tokens; a top-page shortlist can still miss important tokens. Average-key scores do not inherit this bound.

With $K_1 = 128$ and $\beta = .125$, near and far receive 16 and 112 L1 pages. Regional reservation allocates access, not a quality guarantee: a far page can displace a useful near page. A later token quota cannot recover a token after L1 discarded its page. This distinguishes regional

L1 screening from splitting one global shortlist after selection.

For each KV head h, the intended regional support consists of valid tokens in retained pages. The example below assumes this support can fill each quota. For the supported budget domain $K \geq |\mathcal{P}_t|$, the middle cap is $K_{\text{mid}} = K - |\mathcal{P}_t|$. In the sufficient-length configuration derived from the implementation, a per-head total cap $K = 1024$ leaves 892 nominal middle-token slots (Supplementary Fig. S2). In the standard configuration with sufficient eligible candidates, $\gamma = .625$ acts on near-page capacity, yielding $16 \cdot 64 \cdot .625 = 640$ near tokens and 252 far slots. The selected configuration protects four prefix and 128 recent tokens. Because four prefix tokens occupy only part of a page, far selection can reselect up to four protected positions; 252 is a nominal quota, not a guaranteed count of unprotected far tokens. Scarce-support and zero-tie behavior are detailed in Supplementary Note S7.

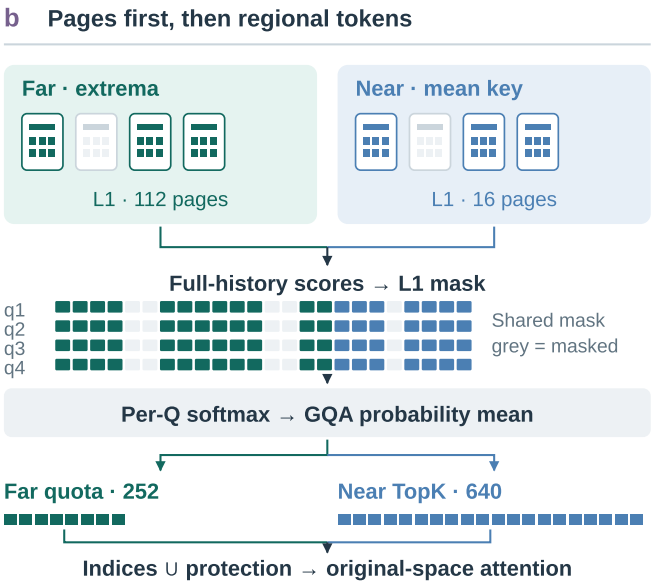


Fig. 2b. Regional screening with full-history L2 scoring. Colored cells indicate eligible candidates, not measured scores. With sufficient eligible capacity, the selected 4/128 prefix/recent configuration has 112/16 far/near pages and nominal 252/640 regional slots, plus 132 protected tokens under cap $K = 1024$. Subpage prefix reselection can consume up to four far slots. Near excludes the protected recent window.

C. GQA selection and original-dimensional attention

With grouped-query attention (GQA), 32 query heads and 8 KV heads give four query heads per KV group. L1 averages their page scores before regional TopK. Ordinary L2 scores the full history for each query, applies the shared L1 mask, and averages the independently normalized query probabilities within each KV group. In this Qwen3 implementation, L2 logits retain the original-head scale $1/\sqrt{128}$ before per-query softmax, including when scoring uses 32 coordinates. This probability aggregation differs from applying softmax to averaged logits. Sharing coordinates therefore preserves query-specific distributions, while full-history scoring means that a smaller page shortlist does not itself reduce L2 dot-product work.

The group's unique selected set $\mathcal{S}_{t,h}$ combines protection and regional winners under the total cap. All queries g in that group use these indices, but recompute their own full-dimensional attention, with head width $d = 128$:

$$o_{h,g} = \sum_{i \in \mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j \in \mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

Merging token indices before this softmax avoids adding separately normalized regional outputs [8]. Retained logits use original KV, but restricting the denominator can change attention output. Selected token indices define a common original-space attention set. Physical gathering and reuse depend on the attention backend and sequence length; the inspected implementation also has a short-sequence dense fallback.

D. Prefill-informed far gating

The optional prefill-informed gate (Fig. 2) asks whether prefill observations can identify layers where repeated far selection can be omitted. Per input and layer, mean prefill far-attention mass below ϵ sets OFF; otherwise far selection stays active. The decision freezes at decode entry and is reused. The feature is disabled by default. In the supported fixed-quota implementation, OFF skips both far scoring stages and their selections, while unused L2 quota refills near candidates without expanding the near-page budget. Representation preparation and sequence-wide score buffers and normalization remain.

Figure 2c shows the gate decision and refill. We implement this extension for a single sequence; Supplementary Note S5 gives the observation boundary and state transitions. Its threshold controls a prefill statistic rather than subsequent attention error. The gate is therefore an approximation policy whose effect on quality and total cost must be evaluated separately from the default ungated method.

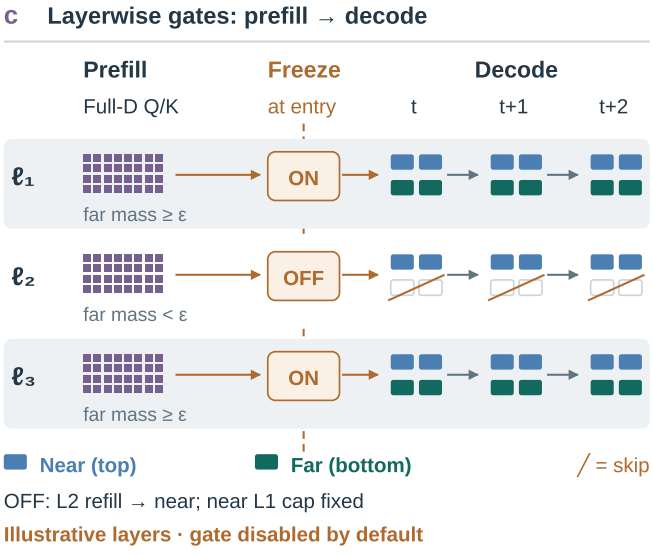


Fig. 2c. Optional far-only control, disabled by default. Full-dimensional causal prefill far-attention mass, averaged over the last ≤ 256 query rows and all query heads, sets one decision per input and layer. Decode entry freezes this state; new inputs reset it. OFF skips far L1/L2 and refills near L2 within unchanged near pages. Layer states are illustrative; paired cells denote L1/L2. This statistic does not bound decoding error.

TABLE I
Selected candidate settings: M denotes experimental configuration; C denotes source-derived behavior. K counts unique final tokens including protection.

Parameter	Setting / provenance
Final token cap K	1024 (M)
Page size; L1 page cap K_1	64; 128 (M)
α, β, γ	.25, .125, .625 (M)
L1/L2 active coordinates	32/32 (C)
Prefix / recent protection	4/128 (M)
Prefill; decode layers	dense (M); all sparse (C)
L2 scoring support	full history before mask (C)

IV. Experimental Setting

The selected Qwen3-8B candidate uses dense prefill and sparse decoding, protecting the first four attention-sink tokens and the most recent 128 tokens (Table I). Their deduplicated union counts toward the total token cap $K = 1024$; content-based selection allocates the remaining slots. The archived five-task scores below retain their original configuration evidence: the shared-subspace run used 128/128 protection and 128-wide, zero-filled L2 keys with 32 active coordinates. Those scores do not validate the selected 4/128 candidate. The selector uses 16-level quantize–dequantize values in floating-point tensors, distinct from packed INT4 arithmetic. Supplementary Note S7 gives configuration and implementation details.

Representation and quality. Dense FullKV retains the full attention support and is the quality reference. The full-dimensional sparse control instead retains sparse selection with full-dimensional scores in both stages. A controlled L1/L2 factorial comparison requires all four modes with fixed scoring rules, protected sets and model settings; the descriptive scores below do not complete that comparison. For sparse baselines, fairness aligns the final unique-token cap K per query-attention row, including protection, on the same model, inputs, decoding phase and dense-layer policy. In our selector, each KV group uses one common selected set; equal K does not imply equal support unions across query heads or equal KV traffic. Internal candidate-page quotas remain method parameters and need not match across methods; their scoring and memory costs are included. A matched comparison must report realized token counts, page rounding and per-query-head versus shared-KV-head selection separately (Supplementary Note S7).

Comparison design. Replay with fixed token indices isolates reuse cost by comparing reuse and reconstruction over the same decode horizon, including initial preparation and per-step maintenance, under the same coordinate map, summary and quantization rules, selected token indices and final attention. To isolate L1 regional allocation, the regional/global comparison instead fixes page size, total L1 cap K_1, representation, protection, L2 refinement and its token quotas, and final cap K; only L1 regional allocation changes. Both configurations use the same page scorer across the history, and any candidate shortage is reflected in realized token counts. These ablation controls do not require competing sparse methods to share K_1. Gate comparisons retain active far selection, far exclusion with near-quota reallocation, and far exclusion without reallocation because removing far can also change the near ranking.

Cost and uncertainty. Complete request time includes representation preparation, maintenance, calibration, selection, KV gathers, attention and synchronization, with overlapping work counted once. TTFT measures time to the first returned token; TPOT averages subsequent return intervals. A task-quality/latency comparison requires the same model, inputs, generation settings and scoring procedure. Uncertainty in task-score differences must account for repeated questions from the same source document. Non-inferiority additionally requires an acceptable-loss margin specified before examining confirmation results. The present analysis reports representation scores and local selector timings separately; it does not estimate a joint end-to-end quality–latency frontier. Supplementary Note S6 records the experiment identities and reproducibility limits.

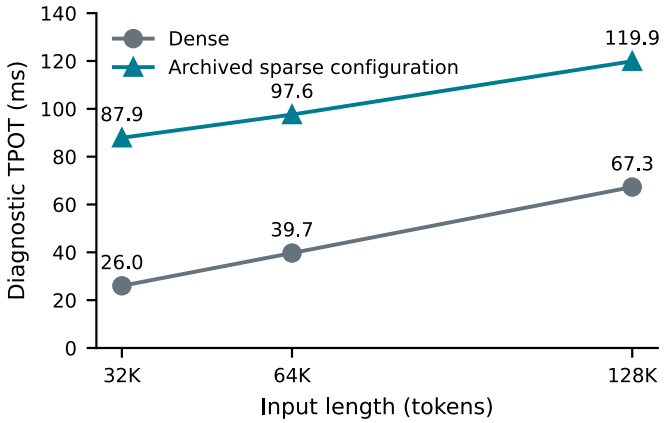


Fig. 3. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including device synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the selected 4/128 candidate. These medians do not measure complete-request E2E or establish a matched quality–cost comparison; provenance limits are in S6.

V. Preliminary Results

Reported configuration scores. Across HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9], the mean scores are 47.588 (shared-subspace configuration), 47.910 (nominal full-dimensional page-scoring variant) and 47.666 (nominal full-dimensional token-scoring variant). Relative to the shared-subspace configuration, the nominal page- and token-scoring variants are higher by 0.322 and 0.078 points; the latter is 0.244 points below the former. Neither variant is verified as a controlled change in scoring dimensionality. Their score differences are therefore descriptive and do not isolate either stage's effect (Supplementary Fig. S4 and Note S7). The archived shared-subspace configuration score does not directly validate the current compact implementation. None is dense FullKV.

The current selected shared-subspace configuration remains unmeasured against dense FullKV under verified matching conditions. Figure 4 instead preserves a historical 128K negative result. The three-configuration means therefore cannot establish preservation of dense-attention quality. Task-level bootstrap summaries describe variation across these five tasks, rather than a document-paired non-inferiority test.

Model-level decode diagnostic. Figure 3 reports TPOT for the archived sparse configuration of 87.9, 97.6 and 119.9 ms at 32K, 64K and 128K, versus dense values of 26.0, 39.7 and 67.3 ms. All three reported sparse times are higher. The synchronized diagnostic excludes tokenization, input transfer and text output; its aggregates do not identify a causal bottleneck. Raw repetition records remain outside the inspected snapshot, and these timings are not bound to the quality runs or the selected 4/128 candidate.

Local selector cost. A separate synthetic experiment uses single-layer, batch-one, single-query QKV at length 64K. Its steady-state timing summaries from two rounds are 1.3502 ms for nominal full-dimensional page scoring, 1.3598 ms for shared-subspace scoring and

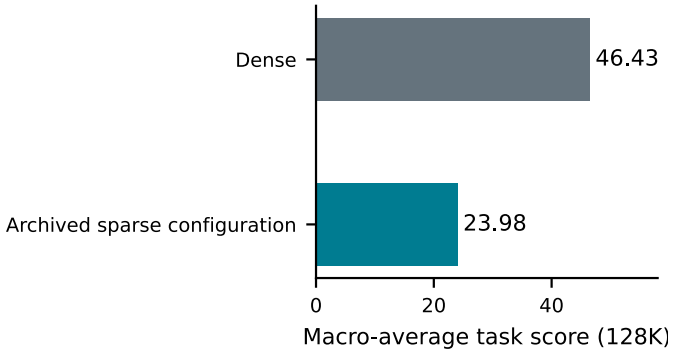


Fig. 4. Author-reported RULER 128K scores: 11 tasks, 100 examples per task and arm. The archived sparse configuration scores 22.45 points below dense FullKV; their macro-average task-score ratio is 51.6%. This historical setting does not validate the selected 4/128 candidate or support sample-paired inference (S6).

1.4634 ms for nominal full-dimensional token scoring. The shared-subspace selector thus has no lower reported time than the nominal page-scoring variant in this setting. This ordering does not identify a limiting stage, and the unresolved sampling-protocol discrepancy prevents a reliable interpretation of small timing differences (Supplementary Note S6). The operator timings are not paired with the LongBench quality runs and do not measure complete-request cost.

Limits of protection and prefill gating. The protection-only configuration scores 26.09 against a 47.91 reference, while using 256 rather than 1024 tokens. The difference therefore combines retrieval-policy and budget effects. Prefill statistics also fail to bound later far-region demand: an observed input–layer pair with prefill far-attention mass below .05 reaches .1678 during decoding. Relative to the reported reference score of 47.910, gate thresholds of .05 and .10 are associated with mean decreases of .034 and .218 points, respectively; three of five tasks decline in each configuration, including HotpotQA decreases of 1.05 and 1.26 points. The larger per-task losses demonstrate that a small average change can conceal task sensitivity. The complete scoring configuration is not identified in these gate summaries, so the results cannot be attributed to the gated shared-subspace method (Supplementary Note S6).

Scope. The results describe configuration scores and the limitations of prefill-based gating on the studied Qwen3 configurations. They do not establish non-inferiority to dense FullKV, the isolated benefit of regional page quotas, or lower complete-request cost. The available 128K RULER summaries [10] do not establish a verified matched comparison for the current compact configuration (Supplementary Note S6). Conclusions are consequently limited to the reported configurations and workloads.

VI. Conclusion

We studied near–far selection with one low-frequency RoPE scoring view shared across both stages. The design reduces scoring width while retaining original-dimensional KV and, in the ordinary implementation, full-history token scoring. Reported configuration-score differences do not isolate stage effects, and the separate synthetic timing study reports no latency advantage for shared-subspace scoring. Whether these representation and allocation choices offer a competitive trade-off therefore depends on task quality and preparation-inclusive request cost under matched final unique token budgets, with acceptable quality degradation specified before evaluation.

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